10.3 Solution: (a) Since $\omega R/c \ll 1$, which is equivalent to

$$\frac{\omega R}{c} = kR = \frac{2\pi R}{\lambda} \ll 1,$$

or $\lambda \gg R$, we can treat the problem as a uniform solid sphere in a constant magnetic field. Also, if the conductivity of the sphere is infinity, there will be no current present. Therefore, we can use magnetic scalar potential to determine the magnetic field around the sphere. Introduce a magnetic scalar potential Φ_m such that

$$\mathbf{B} = -\nabla\Phi_m,$$

and the magnetic scalar potential satisfies the Laplace equation,

$$\nabla^2\Phi_m = 0.$$

Since the magnetic field far from the sphere approaches a constant, we know that

$$\Phi_m = -\mathbf{B}\cdot\mathbf{r}, \qquad \text{as } r \to \infty.$$

Assume that the magnetic field is along the z-axis. Due to the azimuthal symmetry of the problem, the solution to the Laplace equation outside the sphere can be written as

$$\Phi_m = \sum_{\ell=0}^{\infty}\left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}\right)P_\ell(\cos\theta).$$

The boundary condition at infinity, $\Phi_m(r\to\infty) = -Br\cos\theta$, leads to $A_1 = -B$, and $A_\ell = 0$, for $\ell \neq 1$. At the surface of the sphere, we have

$$\left.\frac{\partial\Phi_m}{\partial r}\right|_{r=R} = 0,$$

which gives $B_1 = -\frac{1}{2}BR^3$, and $B_\ell = 0$, for $\ell \neq 1$. Therefore, the magnetic scalar potential outside the sphere is

$$\Phi_m(r,\theta) = -Br\cos\theta - \frac{1}{2}B\frac{R^3}{r^2}\cos\theta,$$

or,

$$\Phi_m = -\mathbf{B}\cdot\mathbf{r}\left(1+\frac{R^3}{2r^3}\right).$$

The magnetic field is then given by

$$\mathbf{B} = -\nabla\Phi_m = \left(\hat{\mathbf{r}}\frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial}{\partial\theta}\right)\left[Br\left(1+\frac{R^3}{2r^3}\right)\cos\theta\right]$$

$$= \hat{\mathbf{r}}B\cos\theta\left(1-\frac{R^3}{r^3}\right) - \hat{\boldsymbol{\theta}}B\sin\theta\left(1+\frac{R^3}{2r^3}\right).$$

(b) From Eq. (8.12), the power loss is

$$\frac{dP_{\text{loss}}}{da} = \frac{\mu\omega\delta}{4}|\mathbf{H}_\parallel|^2 = \frac{\mu\omega\delta}{4\mu_0^2}|\mathbf{B}_\parallel|^2.$$

At the surface,

$$\mathbf{B}_{\parallel} = \hat{\boldsymbol{\theta}} B_\theta = -\hat{\boldsymbol{\theta}} \frac{3}{2} B \sin\theta.$$

Then the total power loss is

$$\begin{aligned} P_{\rm loss} &= \int \frac{dP_{\rm loss}}{da}\, da = R^2 \frac{\mu\omega\delta}{4\mu_0^2} \frac{9}{4} B^2 \int_{-\pi/2}^{\pi/2} \sin\theta\, d\theta \int_0^{2\pi} d\phi\, \sin^2\theta \\ &= \frac{9\mu\omega\delta}{16\mu_0^2} B^2 R^2 \cdot \frac{8\pi}{3} = \frac{3\pi\mu\omega\delta}{2\mu_0^2} B^2 R^2. \end{aligned}$$

The incident power is

$$P_{\rm inc} = \frac{E^2}{Z_0} = \frac{c^2 B^2}{Z_0}.$$

and the absorption cross section is

$$\sigma_{\rm abs} = \frac{P_{\rm loss}}{P_{\rm inc}} = \frac{3\pi R^2\, \mu c\, Z_0\, \omega\delta}{\mu_0^2 c^2}.$$

Since

$$\delta = \sqrt{\frac{2}{\mu\omega\sigma}},$$

we can see

$$\sigma_{\rm abs} \propto \sqrt{\omega}.$$